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Essentials of Data Science
Laboratory - 2304102L - 2026

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

numpyarr...

1 `import numpy as np`

2 `rows,cols=map(int, input().split())`

3 `element=[]`

4 `row=[]`

5 `for _ in range(rows):`

6 `row=list(map(int,input().split()))`

7 `element.extend(row)`

8 `arr1=np.array(element).reshape(rows,cols)`

9

10 `print(arr1)`

11 `print(arr1.ndim)`

12 `print(arr1.shape)`

13 `print(arr1.size)`

14

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3.2.1. Numpy: Matrix Operations

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The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. Addition (`matrix_a + matrix_b`)

2. Subtraction (`matrix_a - matrix_b`)

3. Element-wise Multiplication (`matrix_a * matrix_b`)

4. Matrix Multiplication (`matrix_a . matrix_b`)

5. Transpose of Matrix A

Input Format:

The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.

Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.

2. The result of Subtraction.

3. The result of Element-wise Multiplication.

4. The result of Matrix Multiplication.

matrixOp...

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matrixOperations.py

```
import numpy as np

# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int,
input().split())) for i in range(3)])

print("Enter Matrix B:")
matrix_b = np.array([list(map(int,
input().split())) for i in range(3)])

# Addition
print("Addition (A + B):")
print(matrix_a + matrix_b)

print("Subtraction (A - B):")
print(matrix_a - matrix_b)
# Multiplication (element-wise)
print("Element-wise Multiplication (A * B):")
print(matrix_a * matrix_b)
# Matrix multiplication (dot product)
print("A dot B:")
print(np.dot(matrix_a, matrix_b))

# Transpose
```

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

Horizontal Stacking: Stack the two matrices horizontally (side by side).

Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

The program should first prompt the user to input two 3x3 arrays.

Each array consists of 3 rows, and each row contains 3 space-separated integers.

The user will input the two arrays row by row.

Output Format:

The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.

The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split()))
6                 for i in range(3)])
7
8 print("Enter Array2:")
9 arr2 = np.array([list(map(int, input().split()))
10                 for i in range(3)])
11
12 # Perform horizontal stacking (hstack)
13 ar1=np.hstack((arr1,arr2))
14 print("Horizontal Stack:")
15 print(ar1)
16
17 # Perform vertical stacking (vstack)
18 ar2=np.vstack((arr1,arr2))
19 print("Vertical Stack:")
20 print(ar2)
```

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

Horizontal Stacking: Stack the two matrices horizontally (side by side).

Vertical Stacking: Stack the two matrices vertically (one below the other).

Input Format:

The program should first prompt the user to input two 3x3 arrays.

Each array consists of 3 rows, and each row contains 3 space-separated integers.

The user will input the two arrays row by row.

Output Format:

The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.

The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split()))
6                 for i in range(3)])
7
8 print("Enter Array2:")
9 arr2 = np.array([list(map(int, input().split()))
10                 for i in range(3)])
11
12 # Perform horizontal stacking (hstack)
13 ar1=np.hstack((arr1,arr2))
14 print("Horizontal Stack:")
15 print(ar1)
16
17 # Perform vertical stacking (vstack)
18 ar2=np.vstack((arr1,arr2))
19 print("Vertical Stack:")
20 print(ar2)
```

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4.1 Practice Lab Assignment

3.2.3. Numpy: Custom Sequence Generation 00:58

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

1 import numpy as np

2

3 # Take user input for the start, stop, and step of the sequence

4 start = int(input())

5 stop = int(input())

6 step = int(input())

7 # Generate the sequence using np.arange()

8 sq=np.arange(start,stop,step)

9 # Print the generated sequence

10 print(sq)

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4.1. Practice Lab Assignment

4.1.1 Pandas - series creation and manipu...

3.2.4. Numpy: Arithmetic and Statistical Operations, M... 00:56

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A₁ OR B₁).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     A = np.array(A)
7     B = np.array(B)
8
9     # Arithmetic Operations
10    sum_result = A + B
11    diff_result = A - B
12    prod_result = A * B
13
14    # Statistical Operations
15    mean_A = np.mean(A)
16    median_A = np.median(A)
17    std_dev_A = np.std(A)
18
19    # Bitwise Operations
20    and_result = A & B
21    or_result = A | B
22    xor_result = A ^ B
23
24    # Output results with one space between each element
25    print("Element-wise Sum:", ' '.join(map(str, sum_result)))
```

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3.2.5. Numpy: Copying and Viewing Arrays

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

```
Original array after modifying view: <original_array>
View array: <view_array>
```

- After modifying the copy:

```
Original array after modifying copy: <original_array>
Copy array: <copy_array>
```

```

1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array =original_array.view()
10
11 # Create a copy
12 copy_array =original_array.copy()
13
14
15 # Modify the view
16 view_array[0] = 99
17 print("Original array after modifying view:",
18       original_array)
19 print("View array:", view_array)
20
21 # Modify the copy
22 copy_array[1] = 88
23 print("Original array after modifying copy:",
24       original_array)
25 print("Copy array:", copy_array)

```

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcas...

The given code in the editor takes a single array, array1, as space-separated integers as input from the user. Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching:** Find the indices where search_value appears in array1 and print these indices.
- Counting:** Count how many times count_value appears in array1 and print the count.
- Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

arrayOpe...

Submit

1 import numpy as np

2

3 # Input array from the user

4 array1 = np.array(list(map(int, input().split())))

5

6 # Searching

7 search_value = int(input("Value to search: "))

8 count_value = int(input("Value to count: "))

9 broadcast_value = int(input("Value to add: "))

10

11 # Find indices where value matches in array1

12 indices = np.where(array1 == search_value)

13 # Count occurrences in array1

14 count = np.count_nonzero(array1 == count_value)

15 # Broadcasting addition

16 broadcasted_array = array1 + broadcast_value

17 # Sort the first array

18 sorted_array = np.sort(array1)

19

20

21 print(indices[0])

22 print(count)

23 print(broadcasted_array)

24 print(sorted_array)

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3.2.7. Student Data Analysis and Operations

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.

Find total students: Determine the total number of students in the dataset.

Print all student roll numbers: Extract and print the roll numbers of all students.

Print Subject 1 marks: Extract and print the marks of all students in Subject 1.

Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.

Find maximum marks in Subject 3: Identify the highest marks in Subject 3.

Print all subject marks: Display the marks of all students for each subject.

Find total marks of students: Compute the total marks for each student across all subjects.

Find the average marks of each student: Compute the average marks for each student.

Find average marks of each subject: Compute the average marks for all students in each subject.

Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.

Find average marks of Subject 1 and Subject 3: Compute the

Operatio...

1 import numpy as np

2

3 a = np.loadtxt("Sample.csv", delimiter=',',

4 skiprows=1)

5

6 # 1. Print all student details

7 print("All student Details:\n", a)

8

9 # 2. Print total students

10 print("Total Students:", a.shape[0])

11

12 # 3. Print all student Roll numbers (assuming Roll

13 numbers are in column 0)

14 print("All Student Roll Nos", f"[{'

15 '.join(f'{int(roll)}.' for roll in a[:, 0])}]"

16

17 # 4. Print subject 1 marks (assuming Subject 1

18 marks are in column 1)

19 print("Subject 1 Marks", a[:, 1])

20

21 # 5. Print minimum marks of Subject 2 (assuming

22 Subject 2 is in column 2)

23 print("Min marks in Subject 2", np.min(a[:, 2]))

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